

MASON COUNTY AP, MI, USA

WMO: 726364

Lat: 43.963N

Lon: 86.408W

Elev: 640

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 04883

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	8.7	-3.9	4.6	6.4	0.3	5.7	10.1	25.9	32.6	23.8	33.0	6.2	310	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	83.5	71.8	81.4	70.2	79.9	68.9	74.4	80.7	73.0	78.9	71.4	77.7	10.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	122.9	78.9	71.0	116.8	77.5	69.6	111.6	76.4	38.3	80.8	36.9	79.3	35.7	77.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	19.9	17.8	-3.6	88.6	5.2	2.2	-7.4	90.2	-10.5	91.5	-13.4	92.7	-17.2	94.4
			-4.3	77.5	5.1	2.2	-8.0	79.1	-10.9	80.4	-13.8	81.6	-17.4	83.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	24.7	24.8	32.7	43.7	54.4	63.6	68.1	67.0	60.9	49.3	39.0	30.3
	DBStd	17.58	9.26	8.92	10.50	9.03	8.79	6.68	5.81	5.35	7.50	8.42	7.87	8.11
	HDD50	3406	785	705	548	234	53	1	0	0	5	120	343	612
	HDD65	7060	1250	1125	1002	641	348	102	35	39	167	493	781	1077
	CDD50	2182	0	0	11	44	188	409	561	528	331	97	12	1
	CDD65	362	0	0	1	2	18	60	131	102	43	5	0	0
	CDH74	2707	0	0	4	11	165	509	1000	697	292	29	0	0
	CDH80	370	0	0	0	0	26	79	160	75	29	1	0	0
Wind	WSAvg	7.4	8.6	8.3	7.8	8.4	7.4	6.3	5.9	5.6	6.1	7.5	8.4	8.6
Precipitation	PrecAvg	33.4	2.0	1.5	2.0	2.9	2.6	3.0	2.1	4.0	4.1	3.4	3.1	2.7
	PrecMax	42.3	4.5	3.6	7.1	4.6	5.5	8.3	4.3	9.4	10.7	7.7	7.3	6.4
	PrecMin	24.3	0.4	0.2	0.3	1.0	0.4	0.2	0.6	0.3	0.2	1.0	0.6	0.9
	PrecStd	4.6	1.1	0.9	1.5	1.0	1.4	1.8	1.1	2.2	2.4	1.5	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	52.5	70.0	75.2	82.3	86.1	89.4	84.4	83.6	78.7	64.4	56.4
		MCWB	47.6	47.2	60.5	60.0	68.4	73.5	78.1	72.6	72.0	68.9	56.6	53.6
	2%	DB	43.3	45.0	60.0	70.1	79.0	81.8	83.7	81.8	80.7	72.7	59.1	48.2
		MCWB	39.1	41.5	53.8	56.3	66.4	69.0	72.9	71.4	68.9	64.8	53.0	46.2
	5%	DB	39.4	39.4	53.9	63.9	74.7	79.8	81.3	80.5	77.3	68.6	55.1	43.9
		MCWB	36.8	36.0	47.7	52.3	63.8	67.6	70.9	70.2	68.2	62.0	50.0	40.7
	10%	DB	37.0	36.6	47.9	59.2	70.2	76.5	79.4	77.9	74.9	63.1	52.1	40.7
		MCWB	34.8	33.9	43.0	49.2	59.7	66.2	69.3	68.7	66.7	57.2	47.5	37.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.5	47.4	61.1	64.1	71.9	75.0	78.1	75.7	73.9	70.6	58.5	53.1
		MCDB	48.9	50.8	70.2	73.3	80.9	83.5	86.4	82.1	79.6	77.1	63.1	55.7
	2%	WB	40.1	41.4	54.2	58.2	68.4	71.9	74.9	74.0	71.6	66.5	53.9	46.5
		MCDB	42.5	45.0	58.4	66.6	76.4	79.4	81.6	79.3	77.4	71.2	57.5	47.9
	5%	WB	37.4	37.1	47.8	54.0	65.5	69.8	73.2	72.6	69.8	62.2	50.8	41.3
		MCDB	39.7	39.4	52.6	61.5	72.8	76.3	79.0	78.0	74.5	67.4	54.7	43.7
	10%	WB	34.9	34.1	42.5	50.3	61.7	68.1	71.6	70.9	68.0	58.0	47.8	38.1
		MCDB	37.2	36.4	47.4	58.1	68.1	74.9	77.9	76.4	73.9	62.7	51.9	40.3
Mean Daily Temperature Range	5% DB	MDBR	10.0	12.0	15.3	17.8	19.9	19.2	18.2	18.5	19.1	16.0	12.4	9.2
		MCDBR	12.5	14.9	20.5	25.2	23.1	22.1	18.9	19.4	20.7	20.9	17.6	14.0
		MCWBR	11.3	12.5	15.3	15.4	13.4	12.5	10.9	11.5	12.4	14.6	13.4	13.0
	5% WB	MCDBR	12.6	14.0	19.0	22.0	20.3	18.2	16.1	15.7	16.5	19.1	16.2	14.0
		MCWBR	11.7	12.4	15.3	15.5	13.7	11.9	10.2	10.7	11.6	14.9	13.6	13.1
Clear-Sky Solar Irradiance	taub	0.293	0.312	0.334	0.365	0.396	0.408	0.409	0.401	0.383	0.348	0.316	0.289	
	taud	2.436	2.370	2.395	2.336	2.317	2.325	2.329	2.351	2.384	2.469	2.498	2.485	
	Ebn at Noon	266	279	287	285	278	274	272	271	266	263	255	257	
	Edh at Noon	23	30	33	38	40	40	39	37	33	27	22	20	
All-Sky Solar Radiation	RadAvg	351	628	1120	1500	1807	1987	2019	1714	1342	810	452	279	
	RadStd	42	72	116	141	157	132	91	108	114	96	66	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page